

A Market-Based Analysis of Critical Nodes in Power Systems

Martin Bichler

*School of Computation, Information and Technology
Technical University of Munich
Garching, Germany
bichler@cit.tum.de*

Teodora Dobos

*School of Computation, Information and Technology
Technical University of Munich
Garching, Germany
dobos@cit.tum.de*

Abstract—Assessing power system resilience requires identifying disruptions causing large economic damage after optimal redispatch. We model economic node criticality as a bilevel attacker–defender problem where an attacker disrupts nodes, and a system operator responds with minimum-cost DC optimal power flow. Because such interdiction models scale poorly, we develop fast screening methods for realistic systems. We distinguish between generator-node interdictions, which remove generation capacity but preserve topology, and node interdictions, which additionally eliminate demand and connected lines. These disruption types differ fundamentally. Generator-node interdictions act as right-hand-side perturbations, so cost impacts are well approximated by dual variables from the baseline dispatch. Conversely, node interdictions alter topology and constraint matrix, rendering marginal dual information unreliable; flow-based metrics perform better. Using a 300-node European network from PyPSA-Eur, we show that linear programming-based heuristics closely approximate optimal solutions. Our results imply that resilience assessment must separate marginal scarcity effects from structural network fragmentation.

Index Terms—Electricity Markets, Energy Security and Resilience, Node Criticality

I. INTRODUCTION

Power transmission grids are increasingly operated close to their physical limits due to rising renewable penetration, higher cross-border exchanges, and reduced operational margins [1]. To maintain reliability, transmission system operators (TSOs) enforce the $N - 1$ security criterion by embedding contingency constraints into a security-constrained DC optimal power flow (SC-DCOPF). Since power systems contain thousands of potential outages, explicitly modeling all contingencies is computationally intractable, and TSOs rely on contingency filtering. Approaches implemented in practice identify relevant outages based on their physical influence on the network, such as power flow or voltage influence factors [2]–[4]. These procedures therefore identify operationally severe contingencies.

However, operational severity does not necessarily coincide with economic severity. Following a contingency, the TSO applies remedial actions such as redispatch to restore a secure operating state. The associated cost reflects the need to activate more expensive generation and reroute power flows across the network. Consequently, outages with large physical influence may have limited economic impact, while economically

critical disruptions may remain undetected by physics-based screening metrics.

We therefore evaluate criticality through the cost of optimal corrective dispatch, which admits an economic interpretation corresponding to nodal market clearing under modified operating conditions. We define k critical nodes as those whose removal maximizes post-disruption redispatch cost under optimal operator response. This can be formalized as an attacker–defender bilevel interdiction problem in which an upper-level attacker selects nodes to disrupt and a lower-level operator performs minimum-cost dispatch [5]. This problem can be reformulated as a mixed-integer linear program (MILP) [6], [7], but it scales poorly to large networks and large interdiction budgets k , as the MILP contains numerous binary variables and linearization constraints. Consequently, the literature has followed two main directions: (i) scalable solution methods for the bilevel reformulation that sacrifice optimality guarantees [8], [9], and (ii) fast heuristics and screening approaches that approximate high-impact contingencies [10]–[14] or restrict attacker capabilities (e.g., spatial or topological constraints) to generate credible attack scenarios [15].

This paper contributes to the second direction. We consider two disruption mechanisms: *generator-node interdictions*, where generators at selected nodes are removed, and *node interdictions*, where an entire node together with all connected generation, load, and transmission lines is removed from the grid. To address the limited scalability of the interdiction MILP, we develop LP-based proxy scores and an iterative selection algorithm that derive node criticality directly from the primal and dual variables of the baseline nodal market clearing solution. The proposed methods require only a small number of LP solves and are therefore suitable for screening large networks across many operating states.

We evaluate the methods on a 300-node European network derived from PyPSA-Eur [16]. Our results show that economic criticality depends fundamentally on the disruption mechanism. Generator-node interdictions enter the dispatch problem as right-hand-side perturbations, so the resulting cost increase is well approximated by dual variables of the baseline dispatch solution. In contrast, node interdictions alter the constraint matrix, making marginal price signals unreliable. Here, flow-based structural metrics perform better. These distinct

disruption mechanisms reveal a structural distinction between scarcity-driven and connectivity-driven criticality and imply different screening metrics and resilience planning strategies.

II. METHODOLOGY

A. DCOPF-LP Model

To obtain a reference operating point, we compute a baseline nodal market clearing solution using a linear DC optimal power flow approximation (DCOPF). The model minimizes total system cost consisting of generation cost and the cost of unmet demand (Value of Lost Load, VOLL):

$$\begin{aligned}
& \min_{y, \text{shed}, f, \alpha} \sum_{g \in G} c_g y_g + \sum_{v \in V} \text{VOLL}_v \text{shed}_v \quad (\text{DCOPF-LP}) \\
& \text{s.t.} \quad \sum_{g \in G(v)} y_g + \text{shed}_v - \sum_{(v,w) \in \delta^+(v)} f_{(v,w)} \\
& \quad + \sum_{(u,v) \in \delta^-(v)} f_{(u,v)} = D_v \quad \forall v \in V \quad (\lambda_v) \\
& y_g \leq \bar{P}_g \quad \forall g \in G \quad (\gamma_g) \\
& f_{(v,w)} - B_{vw}(\alpha_v - \alpha_w) = 0 \quad \forall (v,w) \in A \quad (\eta_{(v,w)}) \\
& f_{(v,w)} \leq \bar{F}_{vw} \quad \forall (v,w) \in A \quad (\mu_{(v,w)}^+) \\
& -f_{(v,w)} \leq \bar{F}_{vw} \quad \forall (v,w) \in A \quad (\mu_{(v,w)}^-) \\
& \alpha_{R^+} = 0 \quad (\sigma) \\
& y_g \geq 0, \text{shed}_v \geq 0, \quad f_{(v,w)}, \alpha_v \in \mathbb{R}. \quad (7)
\end{aligned}$$

Dual variables are shown next to each constraint.

B. Generator-Node Interdiction Model

We next study disruptions that remove generation capacity at selected nodes while preserving network topology.

In the *generator-node interdiction (GNI) model*, all generators located at nodes in a set $S \subseteq V$ are removed from service, while load and transmission lines remain operational. We represent interdictions using binary variables $\text{fail}_v \in \{0, 1\}$, where $\text{fail}_v = 1$ indicates that node v is interdicted. The corresponding interdicted set is $S = \{v \in V : \text{fail}_v = 1\}$.

1) *Interdicted Dispatch Problem*: Given a fixed interdiction vector fail , the system operator redispatches generation to restore feasibility at minimum cost. Generators located at interdicted nodes cannot produce power:

$$y_g \leq \bar{P}_g(1 - \text{fail}_{v(g)}) \quad \forall g \in G. \quad (8)$$

The resulting optimization problem is called the *Interdicted Dispatch Problem (IDP)* and its optimal value is

$$C^{\text{GNI}}(\text{fail}) = \min_{y, \text{shed}, f, \alpha} \sum_{g \in G} c_g y_g + \sum_{v \in V} \text{VOLL}_v \text{shed}_v \quad (9)$$

$$\text{s.t.} \quad (1), (3) - (8). \quad (10)$$

The value $C^{\text{GNI}}(\text{fail})$ quantifies the economic impact of removing the nodes $S = \{v : \text{fail}_v = 1\}$.

2) *Bilevel k -Node Interdiction Formulation*: The attacker's goal is to select k nodes whose removal maximizes the operator's minimum dispatch cost. This leads to a bilevel attacker-defender problem, in which the upper level is represented by the attacker and the lower level corresponds to the defender who solves the IDP:

$$\begin{aligned}
& \max_{\text{fail}} C^{\text{GNI}}(\text{fail}) \quad (\text{GNI-Bilevel}) \\
& \text{s.t.} \quad \sum_{v \in V} \text{fail}_v = k \quad \text{fail}_v \in \{0, 1\} \quad \forall v \in V. \quad (11)
\end{aligned}$$

This problem is difficult because the attacker's decision changes the feasible region of the defender's optimization problem, creating a hierarchical and nonconvex structure.

3) *Single-Level Reformulation*: Since the lower-level problem is a linear program, the bilevel GNI problem can be reformulated as a single-level MILP by imposing primal feasibility, dual feasibility, and strong duality. The only bilinear terms arise from products between binary interdiction variables and dual variables associated with generator-capacity constraints. These products are linearized using standard Big-M constraints. The full reformulation is given in Appendix B. Importantly, generator-node interdictions modify only the right-hand side of the capacity constraints (8), while the constraint matrix of the DCOPF-LP remains unchanged. Hence, GNI constitutes a pure right-hand-side perturbation of an LP.

C. Node Interdiction Model

We now generalize the disruption model. While GNI removes only generation capacity, the *node interdiction (NI) model* removes the entire bus from the network, including generators, demand and incident transmission lines.

1) *Interdicted Dispatch Problem*: To model node removal, we modify the DCOPF-LP constraints so that interdicted nodes neither inject nor withdraw power and no longer participate in power flows. Besides integrating the generation constraints (8), we modify the nodal balance constraints (1) as follows

$$\begin{aligned}
& \sum_{g \in G(v)} y_g + \text{shed}_v - \sum_{(v,w) \in \delta^+(v)} f_{(v,w)} + \sum_{(u,v) \in \delta^-(v)} f_{(u,v)} \\
& = D_v(1 - \text{fail}_v) \quad \forall v \in V \quad (\tilde{\lambda}_v), \quad (12)
\end{aligned}$$

Also, when $\text{fail}_v = 1$, it must hold $\text{shed}_v = 0$ and $\alpha_v = 0$:

$$\text{shed}_v \leq D_v(1 - \text{fail}_v) \quad \forall v \in V \quad (\pi_v) \quad (13)$$

$$\alpha_v \leq M^\alpha(1 - \text{fail}_v) \quad \forall v \in V \quad (\phi_v^+) \quad (14)$$

$$-\alpha_v \leq M^\alpha(1 - \text{fail}_v) \quad \forall v \in V \quad (\phi_v^-) \quad (15)$$

Furthermore, transmission lines connected to an interdicted node are taken out of service by replacing (4)–(5) with

$$f_{(v,w)} \leq \bar{F}_{vw}(1 - \text{fail}_v) \quad \forall (v,w) \in A \quad (\tilde{\mu}_{(v,w)}^+) \quad (16)$$

$$-f_{(v,w)} \leq \bar{F}_{vw}(1 - \text{fail}_v) \quad \forall (v,w) \in A \quad (\tilde{\mu}_{(v,w)}^-) \quad (17)$$

$$f_{(v,w)} \leq \bar{F}_{vw}(1 - \text{fail}_w) \quad \forall (v,w) \in A \quad (\tilde{\mu}_{(v,w)}^{+w}) \quad (18)$$

$$-f_{(v,w)} \leq \bar{F}_{vw}(1 - \text{fail}_w) \quad \forall (v,w) \in A \quad (\tilde{\mu}_{(v,w)}^{-w}). \quad (19)$$

2) *Single-Level Reformulation via Strong Duality*: The NI bilevel problem is reformulated similarly to the GNI case by imposing primal feasibility, dual feasibility, and strong duality. The resulting MILP is presented in Appendix C.

D. Node Criticality Heuristics

Solving the MILP reformulations becomes computationally expensive for large networks or large interdiction budgets k . We therefore derive heuristic node criticality scores directly from the optimal baseline DCOPF-LP solution. These methods require only a small number of LP solves and are suitable for large-scale screening problems.

Let $(y^*, \text{shed}^*, f^*, \alpha^*)$ denote the optimal solution of DCOPF-LP and let $\lambda^*, \gamma^*, \mu^{+*}, \mu^{-*}$ be corresponding dual variables. The dual variable λ_v^* represents the locational marginal price (LMP) at node v , while γ_g^* is the shadow price of generator capacity and indicates the marginal value of additional capacity at generator g . Similarly, $\mu_{(v,w)}^{+*}$ and $\mu_{(v,w)}^{-*}$ denote the shadow prices of the transmission capacity constraints and reflect congestion on line (v, w) . Each heuristic assigns a score κ_v to every node and approximates the interdiction set by selecting the top k nodes with highest scores.

The following scores capture economic and physical mechanisms through which node removals increase system cost.

1) *Dispatch Volume Score (DVS)*: This score measures the total baseline dispatch at node v .

$$\kappa_v^{\text{DVS}} = \sum_{g \in G(v)} y_g^*. \quad (20)$$

2) *PTDF Score (PTDFS)*: This score captures the impact of generation at node v on potentially congested lines. It combines installed capacity with network sensitivity and residual line margins. Let $m_l := \bar{F}_l - |f_l^*|$ denote the residual margin on line l , and let PTDF_{lv} be the power transfer distribution factor on line l for a +1 MW injection at node v and a -1 MW withdrawal at the slack node. Then we define

$$\kappa_v^{\text{PTDFS}} = \left(\sum_{g \in G(v)} \bar{P}_g \right) \sum_{l \in A} \frac{|\text{PTDF}_{lv}|}{m_l + \epsilon}, \quad (21)$$

where $\epsilon > 0$ is a small constant set to 10^{-6} to avoid division by zero. Intuitively, PTDFS emphasizes nodes whose removal is likely to shift flows onto already stressed transmission lines.

3) *Load-Weighted LMP (LWLMP)*: This score captures the economic importance of demand at a node: high prices indicate scarcity and high demand amplifies system-wide impact.

$$\kappa_v^{\text{LWLMP}} = \lambda_v^* D_v. \quad (22)$$

4) *Rent-Weighted Dispatch Score (RWDS)*: This score measures realized generator rents, where rent is defined as the price-cost margin $(\lambda_v^* - c_g)$. By weighting this margin with the dispatched output, the scores captures the economic value generated at node v in the baseline solution.

$$\kappa_v^{\text{RWDS}} = \sum_{g \in G(v)} (\lambda_v^* - c_g) y_g^*. \quad (23)$$

5) *Gamma-Capacity-Congestion Score (GCCS)*: This score captures both generation scarcity and local congestion effects. The shadow price γ_g^* reflects the marginal value of generator capacity, while $\mu_{(v,w)}^{+*}$ and $\mu_{(v,w)}^{-*}$ indicate binding transmission constraints. Nodes with high scores host scarce generation and/or are electrically located near congested lines.

$$\kappa_v^{\text{GCCS}} = \sum_{g \in G(v)} \gamma_g^* \bar{P}_g + \sum_{(v,w) \in \delta(v)} \bar{F}_{vw} (\mu_{(v,w)}^{+*} + \mu_{(v,w)}^{-*}). \quad (24)$$

The previous scores evaluate nodes independently using a single baseline operating point. However, interdictions interact: removing one node changes congestion patterns and alters the importance of remaining nodes. To capture these compounding effects, we introduce an iterative selection method.

6) *Iterative-GCCS*: Iterative-GCCS sequentially updates the baseline information after each interdiction. Starting from the initial clearing solution, it computes GCCS scores, selects the highest-scoring non-selected node, resolves the IDP for the updated interdiction set, and repeats until k nodes are selected. This captures interaction effects between interdictions while requiring only $k + 1$ LP solves.

III. RESULTS

A. Experimental Setup

1) *Data*: We evaluate the proposed methods using the PyPSA-Eur framework [16] for all countries in the Single Day-Ahead Coupling region. The resulting network contains 300 nodes. To capture seasonal variability, we consider 10 days from 2013, namely the first day of each month from March to December, each modeled with 24 hourly periods.

2) *Parameters*: For both generator-node and node interdictions, and for budgets $k \in \{1, 5, 10, 20, 30, 50\}$, we solve the corresponding single-level MILP reformulation of the k -node interdiction problem for each hourly period. This yields the optimal interdicted node set S^{OPT} of cardinality k . Following [17], we set $VOLL_v = 500$ EUR/MWh for all nodes v , which exceeds the marginal cost of any generator in the system. All methods were implemented in Python 3.11 using Gurobi, and a time limit of one hour was imposed for each MILP solve.

3) *Metrics*: We consider the following metrics and report quantities averaged over all hourly periods.

a) *Economic Performance*: To evaluate how well a heuristic h approximates S^{OPT} , we solve the IDP using the heuristic node set S^h and compare the resulting cost C^h with the optimal cost of S^{OPT} , C^{OPT} . We report the mean percentage efficiency loss

$$\ell_{\text{pct}}(h, k) = \frac{100}{|I_{h,k}|} \sum_{i \in I_{h,k}} \frac{C^{\text{OPT},(i)} - C^{h,(i)}}{C^{\text{OPT},(i)}}, \quad (25)$$

where $I_{h,k}$ denotes the set of all evaluated instances (day-hour combinations) for heuristic h and budget k .

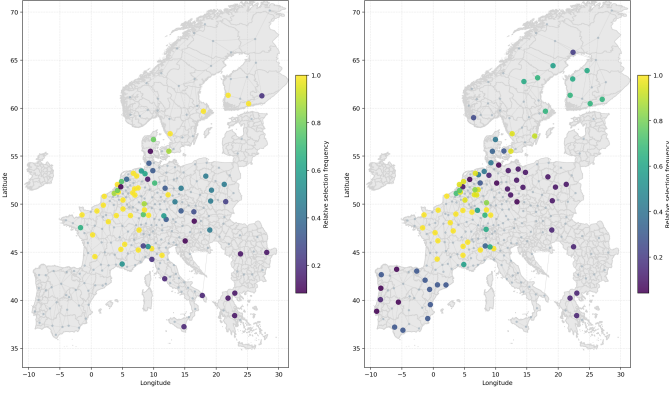


Fig. 1: Generator-Node Interdiction: Node Relative Selection Frequency for July 1 (left) and December 1 (right) computed for $k = 50$.

b) Interdiction set similarity: We measure how closely a heuristic h reproduces the optimal node set S^{OPT} using the Jaccard index, which takes values in $[0, 1]$ with 1 indicating identical sets:

$$J(S^{\text{OPT}}, S^h) = \frac{|S^{\text{OPT}} \cap S^h|}{|S^{\text{OPT}} \cup S^h|}, \quad (26)$$

c) Runtime: We additionally report average computational runtime to assess the scalability of the methods.

B. Generator-Node Interdiction Results

a) Spatial Patterns: Figures 1 illustrate, for July 1 and December 1, the relative frequency with which each node appears in the optimal interdiction set for $k = 50$ across all hourly periods. For both days, the nodes with the highest relative selection frequency are located in France, which also holds for all days we considered. However, the detailed geographic distribution varies with the operating conditions, indicating that while certain regions are structurally critical, the exact set of vulnerable nodes remains time-dependent.

b) Economic Performance: Figure 2 reports the mean percentage efficiency loss of each heuristic. For all heuristics that rely on dual variables (i.e., all except for DVS and PTDFS), the mean percentage efficiency loss monotonically increases with k . This reflects the interaction effects between interdicted nodes: single-node interdictions correspond to small right-hand side perturbations of the constraint matrix of DCOPT-LP Model and thus can be approximated well using information from the baseline solution. However, larger interdiction sets jointly modify system constraints and reduce the accuracy of marginal information. Among the dual-based methods, Iterative-GCCS leads to the lowest efficiency loss, indicating that repeatedly updating dual information captures interaction effects between nodes. This method also achieves the slowest degradation as k increases, suggesting that sequential updating stabilizes the ranking under large perturbations.

Although DVS performs poorly overall, its efficiency loss varies only moderately with k , suggesting that production levels provide a coarse but consistent approximation of the

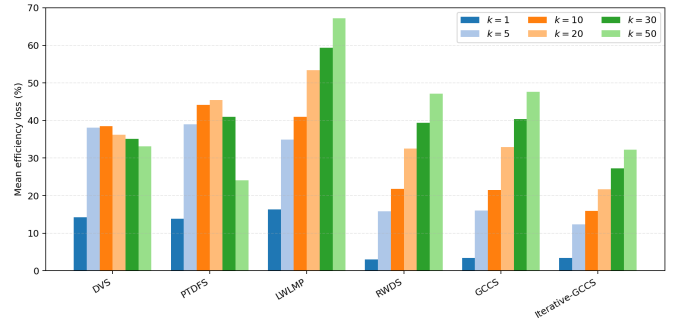


Fig. 2: Generator-Node Interdiction: Mean Percentage Efficiency Loss.

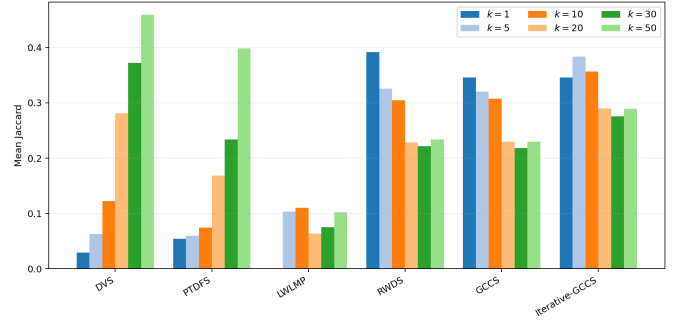


Fig. 3: Generator-Node Interdiction: Mean Jaccard Index.

optimal interdiction sets. Interestingly, for $k = 50$, PTDFS leads to the best result, which shows that as k increases, methods that directly model congestion are superior. Overall, LWLMP - a method that only relies on locational marginal prices and load - performs worse than the other methods.

c) Interdiction Set Similarity: Figure 3 illustrates the mean Jaccard index. Methods RWDS, GCCS, and Iterative-GCCS exhibit both low efficiency loss and high set similarity. However, a high Jaccard index does not necessarily imply good economic performance: DVS attains substantial overlap with the optimal set for $k \in \{30, 50\}$ but still produces large efficiency losses. This suggests that missing a small number of highly critical nodes can dominate system costs even when most nodes are correctly identified.

d) Runtime: Table I shows the average runtime for each method. Recall that to compute the optimal solution, an MILP is solved. All heuristics, except for Iterative-GCCS, solve a single LP, while Iterative-GCCS requires solving $k + 1$ LPs. Thus, OPT leads to the highest average runtime, followed by Iterative-GCCS. DVS, LWLMP, RWDS and GCCS achieve identical average runtime (0.08s), while PTDFS is slightly slower due to the PTDF computation (0.23s).

C. Node Interdiction Results

Unlike GNI, node interdiction changes topology and removes both supply and demand. Economic damage is therefore mainly driven by the modified grid's ability to reroute power.

TABLE I: Generator-Node Interdiction: Average Runtime (s)

Method	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 30$	$k = 50$
OPT	3.68	5.80	24.52	55.23	234.36	159.56
PTDFS	0.23	0.23	0.23	0.23	0.23	0.23
Iterative-GCCS	0.17	0.61	1.16	2.27	3.38	5.64
Others	0.08	0.08	0.08	0.08	0.08	0.08

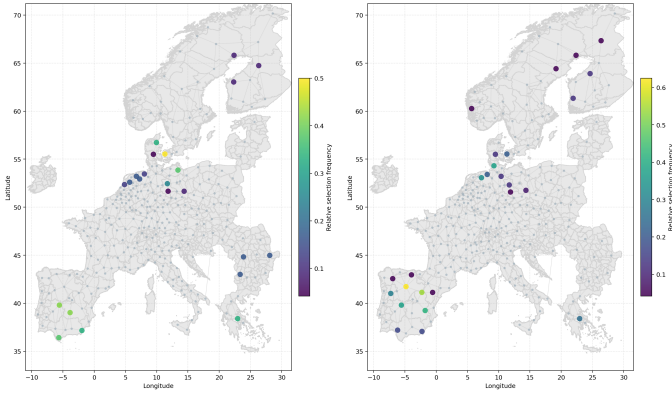


Fig. 4: Node Interdiction: Node Relative Selection Frequency for July 1 (left) and December 1 (right) computed for $k = 5$.

For budgets $k \in \{10, 20, 30, 50\}$, no incumbent MILP solution was obtained within the one-hour time limit.¹ We therefore report node-interdiction results only for $k \in \{1, 5\}$.

a) Spatial Patterns: Figures 4 show the relative selection frequency of nodes for $k = 5$ in the optimal interdiction solution. In contrast to generator-node interdictions, the spatial distribution of selected nodes is more dispersed across the network. This pattern indicates that node interdictions primarily target connectivity rather than local production. Instead of removing economically valuable generators, the attacker seeks to fragment the network into weakly connected zones, limiting power exchange across regions.

b) Economic Performance: Figure 5 shows that all heuristics achieve small efficiency losses, below 3% for $k = 1$ and below 7% for $k = 5$. This contrasts with generator-node interdictions, where performance differences between heuristics were large even for small budgets k . Because a node interdiction removes both supply and demand, the system imbalance remains limited and costs arise mainly when power cannot be rerouted through the modified transmission network. Hence, economic damage is determined by feasibility of power transfer rather than marginal production scarcity.

This mechanism explains the ranking of heuristics. Methods based on marginal economic information (LWLMP, RWDS, GCCS, Iterative-GCCS) perform worst because dual variables reflect marginal values at the baseline operating point and therefore become unreliable after topology changes. Conversely, DVS and PTDFS achieve the lowest losses as they capture volumetric importance and flow redistribution potential. The strong performance of PTDFS indicates that critical nodes are those whose removal forces large flows onto few remaining corridors, independently of local prices.

¹Even with a three-hour limit, a feasible solution was not consistently found.

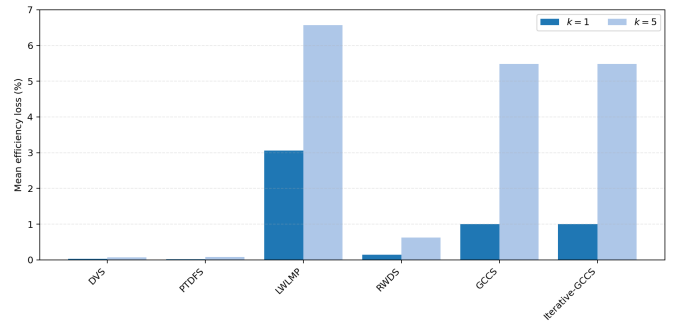


Fig. 5: Node Interdiction: Mean Percentage Efficiency Loss.

c) Interdiction Set Similarity: Despite the small efficiency losses, the overlap between heuristic and optimal node sets is almost zero. For $k = 1$, only PTDFS attains a non-zero mean Jaccard index (0.03), while all other methods yield no overlap. For $k = 5$, similarity remains below 0.04 for all methods. The combination of low mean percentage efficiency losses and low set similarity indicates that comparable damage can be achieved by different node combinations.

d) Runtime: For $k = 1$ and $k = 5$, OPT could be computed in 7.23 and 977.21 seconds, respectively. Iterative-GCCS needed 0.29 seconds for $k = 1$ and 1.06 seconds for $k = 5$. Thus, obtaining the exact interdiction set requires more computation while yielding only marginal economic improvement over heuristic selections.

IV. CONCLUSION

We presented a market-based analysis of economically critical nodes in power systems using an attacker-defender interdiction formulation of DC optimal power flow. Our main contribution is a set of scalable heuristics that approximate optimal interdiction solutions using primal and dual information from the baseline market clearing solution.

Our results show that the effectiveness of different heuristics depends fundamentally on how disruptions enter the dispatch problem. Generator-node interdictions constitute right-hand-side perturbations of the dispatch problem and leave the constraint matrix unchanged. Accordingly, dual variables provide valid marginal sensitivity information. In contrast, node-interdictions modify network topology and alter the constraint matrix, so baseline dual sensitivities lose validity and flow-based structural indicators perform better.

From a practical perspective, our results indicate that large-scale resilience assessment can rely on fast heuristics rather than exact interdiction optimisation. More importantly, the results imply that resilience planning should distinguish between scarcity-driven and connectivity-driven disruptions, as they require different mitigation strategies: protecting high-value generators in the former case and safeguarding critical transmission corridors in the latter.

ACKNOWLEDGMENT

This project was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Grant No. BI 1057/10.

REFERENCES

- [1] K. Janda, J. Málek, and L. Rečka, "Influence of renewable energy sources on transmission networks in central europe," *Energy Policy*, vol. 108, pp. 524–537, 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0301421517303737>
- [2] "Methodology for assessing the relevance of assets for outage coordination," Agency for the Cooperation of Energy Regulators (ACER) / ENTSO-E, ACER Decision on RAOCM: Annex I, June 2019, methodology established under Article 84 of Commission Regulation (EU) 2017/1485 (SO Regulation). [Online]. Available: https://eeepublicdownloads.entsoe.eu/clean-documents/nc-tasks/EBGL/SOGL_A84.1_190621__ACER%20Decision%20on%20RAOC_Annex%20I.pdf
- [3] F. Capitanescu, M. Glavic, D. Ernst, and L. Wehenkel, "Contingency filtering techniques for preventive security-constrained optimal power flow," *Power Systems, IEEE Transactions on*, vol. 22, pp. 1690 – 1697, 12 2007.
- [4] A. Abulwafa, F. Aboul, and S. Nasser Fahmy, "Power system security assessment under n-1 and n-1-1 contingency conditions," *International Journal of Scientific and Engineering Research*, vol. 12, pp. 1854–1863, 11 2019.
- [5] J. Salmeron, K. Wood, and R. Baldick, "Analysis of electric grid security under terrorist threat," *IEEE Transactions on Power Systems*, vol. 19, no. 2, pp. 905–912, 2004.
- [6] J. Arroyo and F. Galiana, "On the solution of the bilevel programming formulation of the terrorist threat problem," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 789–797, 2005.
- [7] A. Motto, J. Arroyo, and F. Galiana, "A mixed-integer lp procedure for the analysis of electric grid security under disruptive threat," *IEEE Transactions on Power Systems*, vol. 20, no. 3, pp. 1357–1365, 2005.
- [8] J. Salmeron, K. Wood, and R. Baldick, "Worst-case interdiction analysis of large-scale electric power grids," *IEEE Transactions on Power Systems*, vol. 24, no. 1, pp. 96–104, 2009.
- [9] A. Delgadillo, J. M. Arroyo, and N. Alguacil, "Analysis of electric grid interdiction with line switching," *IEEE Transactions on Power Systems*, vol. 25, no. 2, pp. 633–641, 2010.
- [10] V. M. Bier, E. R. Gratz, N. J. Haphuriwat, W. Magua, and K. R. Wierzbicki, "Methodology for identifying near-optimal interdiction strategies for a power transmission system," *Reliability Engineering & System Safety*, vol. 92, no. 9, pp. 1155–1161, 2007, critical Infrastructures. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0951832006001712>
- [11] J. Arroyo and F. Fernandez Morales, "Application of a genetic algorithm to n-k power system security assessment," *International Journal of Electrical Power & Energy Systems*, vol. 49, pp. 114–121, Jul. 2013.
- [12] M. J. Eppstein and P. D. H. Hines, "A "random chemistry" algorithm for identifying collections of multiple contingencies that initiate cascading failure," *IEEE Transactions on Power Systems*, vol. 27, no. 3, pp. 1698–1705, 2012.
- [13] S. Soltan and G. Zussman, "Quantifying the effect of k-line failures in power grids," in *2016 IEEE Power and Energy Society General Meeting (PESGM)*, 2016, pp. 1–5.
- [14] C. M. Davis and T. J. Overbye, "Multiple element contingency screening," *IEEE Transactions on Power Systems*, vol. 26, no. 3, pp. 1294–1301, 2011.
- [15] K. Sundar, S. Misra, R. Bent, and F. Pan, "Credible interdiction for transmission systems," *IEEE Transactions on Control of Network Systems*, vol. 8, no. 2, pp. 738–748, 2021.
- [16] T. Brown, J. Hörsch, and D. Schlachtberger, "PyPSA: Python for Power System Analysis," *Journal of Open Research Software*, vol. 6, no. 4, 2018. [Online]. Available: <https://doi.org/10.5334/jors.188>
- [17] A. Ratha, J. Kazempour, A. Virag, and P. Pinson, "Exploring market properties of policy-based reserve procurement for power systems," in *2019 IEEE 58th Conference on Decision and Control (CDC)*, 2019, pp. 7498–7505.

APPENDIX A NOMENCLATURE

Indices and Sets

g	Index of generator
v, w	Indices of nodes (buses)

G	Set of generators
V	Set of nodes (buses)
A	Set of lines (arcs)
$N(v)$	Set of adjacent nodes of node v
$G(v)$	Set of generators located at node v

Parameters

c_{gt}	Variable cost of generator g [EUR/MWh]
$v(g)$	Node where generator g is located
$\delta^+(v)$	Set of outgoing arcs for node v
$\delta^-(v)$	Set of ingoing arcs for node v
$\bar{P}_g$	Installed capacity of generator g [MW]
D_v	Load at node v [MW]
$VOLL_v$	Value of lost load at node v [EUR/MWh]
R^*	Reference (slack) node
B_{vw}	Susceptance of line (v, w) [p.u.]
$\underline{F}_{vw}, \bar{F}_{vw}$	Minimum and maximum bounds for flow on line (v, w) [MW]

Decision Variables

$fail_v$	= 1 if node v is interdicted, 0 otherwise
y_g	Production of generator g [MW]
$shed_v$	Load shedding at node v [MW]
$f_{(v,w)}$	Flow on line (v, w) [MW]
α_v	Voltage angle at node v [rad]

APPENDIX B

GENERATOR-NODE INTERDICTION BILEVEL MODEL: MILP REFORMULATION

$$\max \sum_{g \in G} c_g y_g + \sum_{v \in V} VOLL_v shed_v \quad (27)$$

$$\text{s.t.} \quad (11), (10) \quad (28)$$

$$\lambda_{v(g)} - \tilde{y}_g \leq c_g \quad \forall g \in G \quad (29)$$

$$\lambda_v \leq VOLL_v \quad \forall v \in V \quad (30)$$

$$-\lambda_v + \lambda_w + \eta_{(v,w)} - \mu_{(v,w)}^+ + \mu_{(v,w)}^- = 0 \quad \forall (v, w) \in A \quad (31)$$

$$\sum_{(v,w) \in \delta^+(v)} (-B_{vw} \eta_{(v,w)}) + \sum_{(u,v) \in \delta^-(v)} (B_{uv} \eta_{(u,v)}) + \mathbb{1}_{\{v=R^*\}} \sigma = 0 \quad \forall v \in V \quad (32)$$

$$\sum_{g \in G} c_g y_g + \sum_{v \in V} VOLL_v shed_v = \sum_{v \in V} D_v \lambda_v - \sum_{g \in G} \bar{P}_g (\tilde{y}_g - z_g) - \sum_{(v,w)} \bar{F}_{vw} (\mu_{(v,w)}^+ + \mu_{(v,w)}^-) \quad (33)$$

$$z_g \leq \Gamma^{\max} fail_{v(g)} \quad \forall g \in G \quad (34)$$

$$z_g \leq \tilde{y}_g \quad \forall g \in G \quad (35)$$

$$z_g \geq \tilde{y}_g - \Gamma^{\max} (1 - fail_{v(g)}) \quad \forall g \in G \quad (36)$$

$$\lambda_v, \eta_{(v,w)}, \sigma \in \mathbb{R}, \quad \tilde{y}_g, \mu_{(v,w)}^+, \mu_{(v,w)}^-, z_g \geq 0. \quad (37)$$

The variable z_g is defined as $z_g = fail_{v(g)} \tilde{y}_g$, and $\Gamma^{\max}$ is an upper bound on $\tilde{y}_g$. The MILP consists of the attacker's budget

constraint (11), the defender's primal constraints (10), the dual feasibility constraints (29)–(37), the strong duality constraint in the MILP above, and the linearization constraints (34)–(36). This reformulation follows the classical interdiction approach of [7].

APPENDIX C

NODE INTERDICTION BILEVEL MODEL: MILP REFORMULATION

$$\max \quad \sum_{g \in G} c_g y_g + \sum_{v \in V} VOLL_v \text{shed}_v \quad (38)$$

$$\text{s.t.} \quad (11) \quad (39)$$

$$\text{fail}_{R^*} = 0 \quad (40)$$

$$(3), (6), (8), (12) - (19) \quad (41)$$

$$(29) \quad (42)$$

$$\tilde{\lambda}_v - \pi_v \leq VOLL \quad (43)$$

$$-\tilde{\lambda}_v + \tilde{\lambda}_w + \eta_{(v,w)} - \tilde{\mu}_{(v,w)}^{+v} + \tilde{\mu}_{(v,w)}^{-v} - \tilde{\mu}_{(v,w)}^{+w} + \tilde{\mu}_{(v,w)}^{-w} = 0 \quad \forall (v,w) \in A \quad (44)$$

$$\sum_{(v,w) \in \delta^+(v)} (-B_{vw} \eta_{(v,w)}) + \sum_{(u,v) \in \delta^-(v)} (B_{uv} \eta_{(u,v)}) + \mathbb{1}_{\{v=R^*\}} \sigma + \phi_v^+ - \phi_v^- = 0 \quad \forall v \in V \quad (45)$$

$$\sum_{g \in G} c_g y_g + \sum_{v \in V} VOLL_v \text{shed}_v = \sum_{v \in V} D_v \tilde{\lambda}_v - \sum_{g \in G} \bar{P}_g \tilde{y}_g - \sum_{(v,w) \in A} \bar{F}_{vw} (\tilde{\mu}_{(v,w)}^{+v} + \tilde{\mu}_{(v,w)}^{-v} + \tilde{\mu}_{(v,w)}^{+w} + \tilde{\mu}_{(v,w)}^{-w}) - \sum_{v \in V} M^\alpha (\phi_v^+ + \phi_v^-) - \sum_{v \in V} D_v \pi_v + \sum_{g \in G} \bar{P}_g z_g^\gamma + \sum_{v \in V} D_v z_v^\lambda + \sum_{v \in V} D_v z_v^\pi + \sum_{v \in V} M^\alpha (z_v^{\phi^+} + z_v^{\phi^-}) + \sum_{(v,w) \in A} \bar{F}_{vw} (z_{(v,w)}^{\mu+v} + z_{(v,w)}^{\mu-v} + z_{(v,w)}^{\mu+w} + z_{(v,w)}^{\mu-w}) \quad (46)$$

$$\begin{aligned} & \sum_{(v,w) \in A} \bar{F}_{vw} (\tilde{\mu}_{(v,w)}^{+v} + \tilde{\mu}_{(v,w)}^{-v} + \tilde{\mu}_{(v,w)}^{+w} + \tilde{\mu}_{(v,w)}^{-w}) \\ & - \sum_{v \in V} M^\alpha (\phi_v^+ + \phi_v^-) - \sum_{v \in V} D_v \pi_v \\ & + \sum_{g \in G} \bar{P}_g z_g^\gamma + \sum_{v \in V} D_v z_v^\lambda + \sum_{v \in V} D_v z_v^\pi \\ & + \sum_{v \in V} M^\alpha (z_v^{\phi^+} + z_v^{\phi^-}) + \end{aligned} \quad (47)$$

$$\sum_{(v,w) \in A} \bar{F}_{vw} (z_{(v,w)}^{\mu+v} + z_{(v,w)}^{\mu-v} + z_{(v,w)}^{\mu+w} + z_{(v,w)}^{\mu-w}) \quad (48)$$

Linearization constraints similar to (34) – (36). (49)

Constraint (39) is the attacker constraint, constraint (40) ensures that the reference node is never interdicted, constraints (41) enforce primal feasibility, constraints (42)–(45) ensure dual feasibility, while constraint (48) is the strong duality equality. The bilinear products between binary interdiction variables and continuous dual variables are linearized using standard Big-M inequalities, similar to constraints (34)–(36). For every product of the form

$$z = b \cdot x, \quad b \in \{0, 1\}, \quad 0 \leq x \leq X^{\max},$$

we introduce an auxiliary variable z and enforce

$$0 \leq z \leq X^{\max} b, \quad (50)$$

$$z \leq x, \quad (51)$$

$$z \geq x - X^{\max} (1 - b). \quad (52)$$

This linearization is applied to the products

$$\begin{aligned} z_g^\gamma &= \text{fail}_{v(g)} \tilde{y}_g, \quad z_v^\lambda = \text{fail}_v \tilde{\lambda}_v, \quad z_v^\pi = \text{fail}_v \pi_v, \quad z_v^{\phi^\pm} = \text{fail}_v \phi_v^\pm, \\ z_{(v,w)}^{\mu \pm v} &= \text{fail}_v \tilde{\mu}_{(v,w)}^{\pm v}, \quad z_{(v,w)}^{\mu \pm w} = \text{fail}_w \tilde{\mu}_{(v,w)}^{\pm w}. \end{aligned}$$

Variables satisfy

$$\text{fail}_v \in \{0, 1\}, \quad y_g \geq 0, \quad \text{shed}_v \geq 0, \quad \tilde{y}_g, \pi_v, \tilde{\mu}, \phi_v^\pm \geq 0,$$

$$\tilde{\lambda}_v, \eta_{(v,w)}, \sigma \in \mathbb{R}, \quad f_{(v,w)}, \alpha_v \in \mathbb{R}, \quad z \geq 0.$$